expansion of the vector field, and therefore a similar expansion of the Liouvillian operator.

We begin by defining the generator $\hat{\mathcal{L}}$ associated to the propagation of averages under the numerical method for an SDE. We think of this generator as a perturbation of the exact Langevin dynamics generator $\mathcal{L}_{\mathrm{LD}}$, or

$$\hat{\mathcal{L}} = \mathcal{L}_{\mathrm{LD}} + h\mathcal{L}_1 + h^2\mathcal{L}_2 + h^3\mathcal{L}_3 + \mathcal{O}(h^4). \tag{7.16}$$

However, the operators involved are not equivalent to the operators that would arise from any stochastic differential equation (see [403]), so we must give up the direct analogy between the formal series and continuous SDE models (which is available in the deterministic setting). Nonetheless, it is possible to analyze the resulting operator expansion in a formal way.

We shall similarly consider the invariant (or native) distribution $\hat{\rho}$. We write this invariant distribution as a perturbation of the (target) canonical distribution

$$\hat{\rho} = \rho_\beta \left[1 + hf_1 + h^2 f_2 + h^3 f_3 + \mathcal{O}(h^4)\right], \tag{7.17}$$

with each *correction function* $f_k(\boldsymbol{q},\boldsymbol{p})$ satisfying

$$\int_{\mathcal{D}} f_k(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega = 0,$$

ensuring a useful equivalence in the partition function

$$\int_{\mathcal{D}} \hat{\rho}(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega = \int_{\mathcal{D}} \rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega.$$

Our computed average is therefore

$$\langle\phi\rangle_h \overset{\mathrm{def}}{=} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\hat{\rho}(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega.$$

The invariant distribution can, itself, be calculated by a formal expansion in powers of h, although in general this requires the solution of partial differential equations for the terms f_k. In some limiting cases it is possible to write down the solutions explicitly and to obtain, therefore, an understanding of the error induced in an average computed using a numerical method.